

From Master Integrals to Functions: the Differential-Equation Pipeline of the NNLO Double-Real Channel

Lecture notes, full version

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August 24, 2026

Abstract

These notes explain, from first principles, how the 347 master integrals of the double-real channel of $pp \rightarrow h + X$ at NNLO are turned from abstract basis elements of an IBP reduction into explicit analytic functions of the kinematics, expanded in the dimensional regulator ε . The chain of ideas is: masters obey first-order linear differential equations in the kinematic invariants (Section 2); the equations organize into 91 block-triangular *family* systems whose diagonal blocks fall into only 173 equivalence *classes* (Section 3); each class is brought to *canonical* (ε -)form, in which the equations solve themselves order by order in ε as iterated integrals (Sections 4–5); square roots of kinematic polynomials are tamed by rational *charts* or, when no chart exists, by working in the Galois algebra of the roots (Section 6); the general solution is *transported* through kinematic space as words in a dlog alphabet (Section 7); and the remaining integration constants — the *boundary periods* — are pinned down by a small set of physical principles, of which only a handful require computing any new integral (Section 8). A final section explains why the singular behaviour of the solutions near the phase-space endpoints is encoded in the equations themselves and how it is recovered without expanding it away (Section 9). Every concept is defined before use and every mechanism is illustrated on an example small enough to check by hand.

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1 The object of study

1.1 Where the integrals come from

The physical target is the NNLO hard function for single-inclusive hadron production, $pp \rightarrow h + X$, in collinear factorization. The double-real channel is the contribution in which two additional partons are radiated into the final state and integrated over. Its building blocks are therefore *phase-space integrals* of squared tree amplitudes: integrals of rational functions of momenta over the momenta of two unobserved massless particles, subject to on-shell delta functions.

By *reverse unitarity* each on-shell delta function is traded for the discontinuity of a propagator,

$$\delta^{(+)}(p^2) \longrightarrow \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right], \quad (1)$$

which converts every phase-space integral into a (cut) two-loop integral. The point of this trade is purely structural: cut integrals obey the same integration-by-parts (IBP) identities as ordinary

loop integrals, so the entire technology of loop reduction applies. The “cut” lines are bookkept throughout — an integral in which a cut propagator has been cancelled is zero, and this rule propagates through every identity below.

After projection onto the twist-two correlators, the hard functions depend on two dimensionless Mandelstam ratios. In these notes we call them (v, w) , with physical region

$$0 < v, \quad 0 < w, \quad v + w < 1, \quad (2)$$

and everything is computed in dimensional regularization, $d = 4 - 2\varepsilon$. (In the package the same two variables appear as x, y at the amplitude stage and as v, w at the master stage; the letters x, y are reserved at the master stage for *chart* variables, Section 6.)

1.2 What “solving the masters” means

IBP reduction expresses every integral of the problem as a $\mathbb{Q}(v, w, \varepsilon)$ -linear combination of a finite basis of *master integrals*; for the double-real channel there are

$$347 \text{ masters.} \quad (3)$$

The reduction is exact but purely algebraic: it tells us nothing about the *values* of the masters. “Solving” them means producing, for each master $I(v, w; \varepsilon)$, a Laurent expansion

$$I(v, w; \varepsilon) = \sum_{k \geq k_0} \varepsilon^k I^{(k)}(v, w), \quad (4)$$

in which every coefficient $I^{(k)}$ is an explicit, evaluable function — in practice a $\mathbb{Q}$ -linear combination of *iterated integrals* (generalized polylogarithms) of the kinematics with exactly known constant prefactors — to a depth k dictated by where the master enters the cross section. The rest of these notes is the story of how one gets there. The route has five conceptual stations:

1. derive first-order differential equations in (v, w) for the masters (Section 2);
2. organize them into families and classes so the hard work is done once per class, not once per master (Section 3);
3. transform to canonical ε -form, the representation in which the equations integrate themselves (Sections 4, 5, 6);
4. integrate: build the general solution as words in a dlog alphabet (Section 7);
5. fix the integration constants from physics (Section 8) and extract the endpoint behaviour without destroying it (Section 9).

2 From IBP to differential equations

2.1 Why masters obey differential equations

Fix an *integral family*: a maximal set of inverse propagators $D_1, \dots, D_N$ (including the cut ones) such that every scalar product of loop and external momenta is a linear combination of the D_i and the external invariants. The family’s integrals are

$$I_{a_1 \dots a_N}(v, w; \varepsilon) = \int \prod_{\ell=1}^2 \frac{d^d k_\ell}{i\pi^{d/2}} \frac{1}{D_1^{a_1} \dots D_N^{a_N}}, \quad a_i \in \mathbb{Z}, \quad (5)$$

with the cut lines fixed at power one and never cancelled. Two facts combine into the differential equations:

1. **Differentiation stays inside the family.** A derivative ∂_v acting on the integrand can be written, by the chain rule, as a combination of derivatives with respect to external momenta contracted into the loop integrand. Each such derivative raises or lowers propagator powers: it produces integrals $I_{a'_1 \dots a'_N}$ of the same family.
2. **IBP closes the system.** Every integral of the family reduces to the finite master basis $\vec{I} = (I_1, \dots, I_n)^\top$ of that family and of its *subsectors* (families obtained by deleting propagators).

Applying (1) then (2) to each master gives a closed linear system

$$\boxed{\partial_v \vec{I} = A_v(v, w; \varepsilon) \vec{I}, \quad \partial_w \vec{I} = A_w(v, w; \varepsilon) \vec{I}}, \quad (6)$$

with A_v, A_w matrices of rational functions of (v, w, ε) . We package the pair as a *connection* one-form $A = A_v dv + A_w dw$ and write $d\vec{I} = A\vec{I}$.

Example 2.1 (The one-mass bubble, done by hand). *The mechanism in miniature, on the classic one-loop example. Let*

$$B(s; \varepsilon) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{k^2 (k+p)^2}, \quad s = p^2. \quad (7)$$

The family has $N = 2$ propagators, $D_1 = k^2$, $D_2 = (k+p)^2$. Its only nontrivial master is B itself; its subsector (delete D_2) is the massless tadpole, which vanishes. Differentiate:

$$\frac{\partial B}{\partial s} = \frac{1}{2s} p^\mu \frac{\partial B}{\partial p^\mu} = -\frac{1}{2s} \int \frac{d^d k}{i\pi^{d/2}} \frac{2p \cdot (k+p)}{k^2 [(k+p)^2]^2}. \quad (8)$$

Write the numerator in terms of inverse propagators, $2p \cdot (k+p) = (k+p)^2 - k^2 + s = D_2 - D_1 + s$:

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} [I_{1,1} - I_{2,0} + s I_{1,2}], \quad (9)$$

where $I_{1,1} = B$, $I_{2,0} = 0$ (scaleless), and $I_{1,2}$ is reduced by the standard IBP identity $0 = \int d^d k \partial_{k^\mu} [k^\mu / (D_1 D_2)]$, which yields $I_{1,2} = -\frac{d-3}{s} I_{1,1}$. Substituting,

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} [B + (d-3)B] = -\frac{1-\varepsilon}{s} B \quad (d = 4 - 2\varepsilon). \quad (10)$$

A one-line first-order equation whose solution is $B \propto s^{-\varepsilon} s^0 \cdot s^{-1+\dots}$ — more precisely $B = c(\varepsilon) (-s)^{-\varepsilon} / s^0$ up to normalization: the equation fixes the full s -dependence, $B(s) = c(\varepsilon) (-s)^{-\varepsilon} \cdot s^{-(1-\varepsilon)+\varepsilon}$; carrying out the algebra, $B(s) = c(\varepsilon) (-s)^{-\varepsilon}$ with $c(\varepsilon) = \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)^2 / (\varepsilon(1-2\varepsilon)\Gamma(1-2\varepsilon))$ fixed by one direct evaluation (the “boundary constant”). Every ingredient of the full problem — differentiation into the family, IBP closure, a first-order system, one undetermined constant — is already visible here.

2.2 Block-triangular structure

Order the sectors of a family by the number of propagators. A derivative of a master never generates a sector with *more* propagators than appear in the master’s own sector plus what IBP

can produce, and reduction only moves *down* into subsectors. Consequently, if the master vector is ordered subsectors-first, the connection is *block lower triangular*:

$$A = \begin{pmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{b1} & a_{b2} & \cdots & a_{bb} \end{pmatrix}, \quad (11)$$

where each diagonal block a_{ss} couples only the masters of one strongly connected sector group, and the off-diagonal blocks feed lower sectors into higher ones. This shape drives the entire strategy: diagonal blocks are the irreducible analytic content; off-diagonal blocks are “plumbing” that can be attached afterwards (Section 5).

2.3 Integrability

Because $\partial_v \partial_w \vec{I} = \partial_w \partial_v \vec{I}$, the two matrices in Eq. (6) are not independent: they must satisfy the *flatness* (integrability) condition

$$\partial_v A_w - \partial_w A_v + [A_v, A_w] = 0, \quad \text{i.e.} \quad dA - A \wedge A = 0. \quad (12)$$

Flatness is used in three ways in the pipeline: as a stringent correctness check on every derived or transformed connection; as the guarantee that solutions can be continued along *any* path in the (v, w) region with a path-independent result (Section 7); and as a labour saver — once flatness of a source system is certified, the flatness of any exactly gauge-equivalent system follows and need not be recomputed from scratch.

3 Families, blocks, and the 173 classes

3.1 Canonical families

Definition 3.1 (Integral family, canonical family). *An integral family is a set of inverse propagators as above. Two families are equivalent if a loop-momentum redefinition $k_\ell \rightarrow M_{\ell m} k_m + (\text{external})$ with $M \in GL(2, \mathbb{Z})$, $|\det M| = 1$, maps one propagator set bijectively onto the other while sending cut lines to cut lines. Such shifts leave the integration measure invariant, so equivalent families have literally equal integrals up to index relabeling. A canonical family is a representative of an equivalence class, given a stable name CFn .*

This is the “momentum shifting” one half-remembers. Detection does not search over shifts: a family is characterized, up to exactly these shifts, by its Symanzik polynomials, and Pak’s canonical ordering makes equivalence a hash comparison; each detected equivalence is then *verified* by explicitly constructing the shift. At NNLO double-real this merges 1898 raw families into 430 canonical ones, of which 374 carry reduction targets, and of *those* the 91 families listed below actually carry masters.

Because differentiation is family-local (Section 2), the differential-equation problem splits into one independent system per master-carrying canonical family:

$$347 \text{ masters} = \bigsqcup_n \vec{I}_{\text{CFn}}, \quad 91 \text{ systems of dimension 2 to 47.} \quad (13)$$

“Forming the families of differential equations” is therefore not an extra construction: the families were fixed at reduction time, and the DE inherits their boundaries. What has to be *built* per family is the pair (A_v, A_w) , by differentiating each master and reducing, and the certified statement that Eq. (12) holds.

3.2 Blocks and classes

Definition 3.2 (Block). *A block of a family system is a strongly connected component of the dependency graph of its connection: a minimal group of masters that appear in each other’s derivatives. Ordered by sectors, the blocks are exactly the diagonal blocks a_{ss} of Eq. (11). The 91 family systems contain 1119 blocks in total.*

Definition 3.3 (Class). *Two blocks (possibly from different families) are equivalent if their connections are related by a permutation of the basis, optionally composed with the kinematic involution $v \leftrightarrow w$, as an exact rational matrix identity. A class is an equivalence class of blocks; the double-real inventory has 173 classes. Each class carries a representative block and, once solved, a stored ε -form for that representative.*

Why do so many of the 1119 blocks coincide? Physically: a block is the differential system of one subtopology’s masters, and the same subtopology occurs inside many parent families — the momentum-shift merging happened at the level of *complete* families, but sub-diagrams recur far more often than whole diagrams. The involution $v \leftrightarrow w$ is the crossing $t \leftrightarrow u$ of the underlying process, which maps cut diagrams to cut diagrams. Equivalence is decided by exact identity of matrices, so no physics argument is ever *trusted*: a candidate permutation-plus-swap either reproduces the representative entry by entry or the blocks are not identified.

The economics of the class decomposition is the point: instead of canonicalizing 1119 blocks, one canonicalizes 173 representatives — and the many-page labour of Sections 5 and 6 happens 173 times, not 1119 and not 347. The map back (which family rows realize which class, with which permutation and swap) is bookkept in a registry and re-verified when the family is assembled.

4 The canonical (ε -)form and why it solves the equations

4.1 Definition

Definition 4.1 (ε -form, letters, alphabet, residues). *A family (or block) system is in canonical form, or ε -form, if its connection is*

$$\boxed{d\vec{F} = \varepsilon \sum_a R_a d \log \varphi_a(v, w) \vec{F}}, \quad (14)$$

where the R_a are constant rational matrices (the residues) and the φ_a are polynomials (or algebraic functions, Section 6) of the kinematics, free of ε : the letters. The set $\{\varphi_a\}$ is the alphabet. The regulator appears exactly once, as an overall prefactor.

For a real specimen: the certified family CF12 (a 14-master system) has the six-letter alphabet

$$\{v, w, 1-v, 1-w, v+w, 1-v-w\} \quad (15)$$

(written in its chart variables in the record), and every entry of its connection is ε times a $\mathbb{Q}$ -linear combination of $d \log$ ’s of these six polynomials. Harder families have richer alphabets — CF231’s

includes quadratic letters like $1 + 2s + s^2 + y - 6sy + s^2y$ in its chart variables — but the shape (14) is identical.

Three immediate consequences of the shape:

- **Fuchsian singularities.** The connection has only logarithmic (simple-pole) singularities, located exactly on the zero loci $\varphi_a = 0$. No higher poles, no poles hidden at ε -dependent locations.
- **Manifest integrability.** Since each $d \log \varphi_a$ is closed, flatness (12) reduces to the constant-matrix condition $\sum_{a < b} [R_a, R_b] d \log \varphi_a \wedge d \log \varphi_b = 0$ restricted to the algebraic relations among the letters — checked once, on constants.
- **Uniform weight.** Solutions normalized at a point are, order by order in ε , iterated integrals of uniform *transcendental weight* k at order ε^k (weight = number of integrations; $\log$ has weight 1, Li_2 weight 2, π^2 weight 2, $\dots$).

4.2 Why the ε -form integrates itself

Write $\hat{A} = \sum_a R_a d \log \varphi_a$, so $d\vec{F} = \varepsilon \hat{A} \vec{F}$ with $\hat{A}$ independent of ε . Expand $\vec{F} = \sum_{k \geq 0} \varepsilon^k \vec{F}^{(k)}$ and match powers:

$$d\vec{F}^{(0)} = 0, \quad d\vec{F}^{(k)} = \hat{A} \vec{F}^{(k-1)} \quad (k \geq 1). \quad (16)$$

The system has become a *recursion of quadratures*: the zeroth order is constant; every next order is obtained by integrating known functions. Integrating from a base point p_0 along a path γ ,

$$\vec{F}^{(k)}(p) = \vec{F}^{(k)}(p_0) + \int_{\gamma} \hat{A} \vec{F}^{(k-1)}, \quad (17)$$

and unfolding the recursion produces *Chen iterated integrals*

$$\vec{F}(p) = \mathbb{P} \exp\left(\varepsilon \int_{\gamma} \hat{A}\right) \vec{F}(p_0) = \sum_{k \geq 0} \varepsilon^k \sum_{a_1, \dots, a_k} R_{a_k} \cdots R_{a_1} \underbrace{\int_{0 \leq t_1 \leq \dots \leq t_k \leq 1} \prod_i d \log \varphi_{a_i}(\gamma(t_i))}_{\text{a weight-}k \text{ iterated integral}} \vec{F}(p_0). \quad (18)$$

Flatness makes the answer independent of the path within a fixed homotopy class. The analytic problem is thereby *finished* up to two data: the alphabet functions along paths (a bookkeeping problem, Section 7) and the boundary vector $\vec{F}(p_0)$ (a physics problem, Section 8). This, in one sentence, is why we want the ε -form: *it converts a coupled system of PDEs with ε entangled in every coefficient into a triangular tower of explicit integrals whose ε -dependence is purely book-kept by the expansion order.*

Contrast the generic situation: if A depends on ε non-trivially, the orders of the ε -expansion of $\vec{I}$ do not decouple — $d\vec{I}^{(k)}$ involves $\vec{I}^{(k)}$ itself through the ε^0 part of A , so one must solve a genuine coupled ODE system at every order, with no uniform structure and no control over the function space of the answer.

Example 4.2 (A 1×1 warm-up). $dF = \varepsilon d \log w F$ integrates to $F = C w^\varepsilon = C(1 + \varepsilon \log w + \frac{\varepsilon^2}{2} \log^2 w + \dots)$: the recursion (16) generates the logarithm tower of the exact power w^ε . Note the two readings of the same object: expanded, it is a polynomial in $\log w$ at each order; unexpanded, it is a power w^ε whose exponent is the (here 1×1) residue times ε . Both readings are needed later (Section 9).

Example 4.3 (A 2×2 system producing a dilogarithm). *Take the alphabet $\{w, 1 - w\}$ and*

$$d\vec{F} = \varepsilon \left[\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} d \log w + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} d \log(1 - w) \right] \vec{F}, \quad \vec{F}(w_0=0^+) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (19)$$

(boundary chosen regular at $w = 0$; the precise meaning of “value at a singular point” is Section 9). Order by order for the second component F_2 :

$$F_2^{(1)}(w) = \int_0^w d \log(1 - w') F_1^{(0)} = \log(1 - w), \quad (20)$$

$$F_2^{(2)}(w) = \int_0^w \left[d \log(1 - w') F_1^{(1)} - d \log w' F_2^{(1)} \right] = - \int_0^w \frac{dw'}{w'} \log(1 - w') = \text{Li}_2(w), \quad (21)$$

using $F_1^{(k)} = \delta_{k0}$ (the first row is decoupled here). The weight-2 dilogarithm appears exactly as the two-fold iterated integral with the word $(w, 1 - w)$; at order ε^3 the same mechanics produces Li_3 and products, and so on. Every master in the project is, at each order, a $\mathbb{Q}$ -combination of precisely such words in its family’s alphabet.

Remark 4.4 (Existence). *Nothing guarantees a rational ε -form exists for a given block. For the double-real inventory it does exist for every one of the 173 classes — but for some only after a change of variables that rationalizes a square root (Section 6), and for the three-root families the question is precisely the current frontier. The working attitude: the ε -form is a certificate format, and any method whatsoever may be used to find the transformation, as long as the final identity is checked exactly.*

5 Finding the transformation to ε -form

The unknown is an invertible matrix $T(v, w; \varepsilon)$, the *gauge transformation*, such that the transformed connection

$$A' = T^{-1} A T - T^{-1} dT \quad (22)$$

has the form (14). (Check the transformation law: if $d\vec{I} = A\vec{I}$ and $\vec{I} = T\vec{F}$, then $d(T\vec{F}) = dT\vec{F} + T d\vec{F} = AT\vec{F}$, i.e. $d\vec{F} = (T^{-1}AT - T^{-1}dT)\vec{F}$.) The strategy exploits the block triangle (11): first each *diagonal* block (once per class), then the *off-diagonal* completion (once per family), then a final constant-in- (v, w) transformation that concentrates left-over ε -dependence.

5.1 Diagonal blocks: normal forms and balances

For the diagonal blocks two classical ideas suffice (implemented by the CANONICA-style rational ansatz and the Lee/Libra reduction; both are used in the project as *candidate generators*, with exact verification as the only acceptance).

(i) Rational ansatz. Parametrize T as a matrix of rational functions with denominators built from an educated guess of the alphabet and numerators of bounded degree with unknown ε -dependent coefficients; demand (22) take the form (14); solve the resulting linear system for the coefficients. Blunt but complete at fixed degree.

(ii) **Fuchsification and balances.** First reduce all poles of the connection to simple poles (*Fuchsian* form), then adjust residue eigenvalues to be proportional to ε , using *balance transformations*

$$T = 1 + \frac{c}{v - v_1} P \quad (P^2 = P \text{ a projector}), \quad (23)$$

which shift eigenvalues of the residue at v_1 by integers while touching no other singular point. A sequence of balances walks all eigenvalues into the normalized window, after which a constant transformation splits off the overall ε .

Example 5.1 (Removing a double pole by hand). *Consider the scalar equation*

$$\frac{dF}{dw} = \left(\frac{\alpha}{w^2} + \frac{\beta \varepsilon}{w} \right) F. \quad (24)$$

The w^{-2} term is non-Fuchsian. Substitute $F = e^{-\alpha/w} G$ (an “exponential integrating factor”, the scalar shadow of a balance):

$$\frac{dG}{dw} = \frac{\beta \varepsilon}{w} G \quad \implies \quad G = C w^{\beta \varepsilon}. \quad (25)$$

In the multi-master world the analogous statement is that an apparent higher pole sits in a nilpotent or removable direction and a rational T eliminates it; when it does not (a genuine irregular singularity), no ε -form exists in those variables — which for our inventory never happens after the correct chart is chosen.

5.2 Off-diagonal completion: the strip equation

Assume all diagonal blocks are already in ε -form (their class transformations having been inserted), and let the connection be block-triangular with ε -form diagonal blocks $a_{kk} = \varepsilon e_k$ and non-canonical couplings b_{kj} ($j < k$). Complete one row at a time with the unipotent gauge

$$T = 1 + D, \quad D \text{ supported on block row } k, \text{ columns } j < k. \quad (26)$$

Because D is strictly block-triangular, $D^2 = 0$ and $T^{-1} = 1 - D$, so (22) gives *exactly* (no expansion)

$$A' = A + AD - DA - dD. \quad (27)$$

The (k, j) block of the requirement “ A' is in ε -form” reads

$$\boxed{dD_{kj} = \varepsilon(e_k D_{kj} - D_{kj} c_j) + \bar{b}_{kj} - \varepsilon \sum_a K_a d \log \varphi_a,} \quad (28)$$

where c_j is the (already canonical) diagonal block of column j , $\bar{b}_{kj} = b_{kj} - \sum_{j < m < k} D_{km} b_{mj}$ collects the original coupling and the feed-through of previously completed blocks of the same row, and the constant matrices K_a are unknowns on the same footing as D_{kj} : they become the residues of the completed block. Equation (28) is a *linear* PDE for the pair (D_{kj}, K_a) — linearity is what makes the whole completion tractable.

Example 5.2 (A 2×2 completion by hand). *Alphabet $\{v, 1+v\}$, one variable for brevity. Suppose*

$$A = \begin{pmatrix} 0 & 0 \\ 1 & \varepsilon d \log v \cdot 1 \\ 1+v & \end{pmatrix} \quad \text{schematically: } e = d \log v, \quad c = 0, \quad b = \frac{dv}{1+v}. \quad (29)$$

The coupling b carries no ε , so it cannot be a residue term; try D rational in v with K_{1+v} constant:

$$D' = \varepsilon \frac{D}{v} + \frac{1}{1+v} - \varepsilon \frac{K}{1+v}. \quad (30)$$

At order ε^0 : $D'_0 = \frac{1}{1+v}$ would force $D_0 = \log(1+v)$ — not rational. The resolution is the standard one: relax the target by allowing the completed block to keep an ε -independent residue piece only if it is a pure dlog with constant coefficient — here it is ($b = d \log(1+v)$!), so the correct move is $D = 0$, K absorbing at order ε^0 : the block b is already $d \log(1+v)$ with constant residue 1, and a subsequent constant ε -dependent transformation (next subsection) rescales the row to pull the residue into the εR_a shape. The example teaches the two mechanisms in the production system: rational D absorbs what is absorbable; constant residues K_a and the final constant $T(\varepsilon)$ absorb the rest.

5.3 Solving the strip equation over finite fields

For real blocks (D_{kj} up to 4×24 entries of rational functions with numerators of high degree) Eq. (28) is solved by *ansatz plus modular linear algebra*:

1. **Ansatz.** $D_{kj} = N(v, w; \varepsilon) / \prod_a \varphi_a^{m_a}$ with the denominator built from the letters and the numerator a polynomial with unknown coefficients; the support (which monomials) is bounded by a certified degree census of the forcing $\bar{b}_{kj}$.
2. **Sampling.** Substitute many rational points (v_i, w_i) and regulator values ε_j , all reduced modulo a 31-bit prime p : each sample turns (28) into exact linear equations over $\mathbb{F}_p$ for the unknown coefficients. Solve; repeat over several primes and enough samples.
3. **Reconstruction.** Chinese-remainder the per-prime images and lift each coefficient from $\mathbb{Z}/p_1 p_2 \cdots$ back to $\mathbb{Q}$ by rational reconstruction; validate on samples never used in the fit and at a fresh prime.
4. **Exact certificate.** The reconstructed (D, K) is substituted back into (28) *symbolically* (or into the equivalent whole-family identity at the end): only this exact identity, never the statistics of the fit, is the acceptance.

Example 5.3 (Rational reconstruction in one line). *Modulo $p = 101$: suppose a fitted coefficient is $29 \pmod{101}$. The lattice reduction behind rational reconstruction finds the smallest fraction congruent to it: $29 \equiv \frac{1}{7} \pmod{101}$ (since $7 \cdot 29 = 203 = 2 \cdot 101 + 1$). With a second prime, $p = 103$: $\frac{1}{7} \equiv 59 \pmod{103}$; CRT of $(29, 59)$ over $101 \cdot 103$ reconstructs to $\frac{1}{7}$ again, and agreement of the lift across growing moduli (plus a fresh-prime check) is the statistical evidence — the proof remains the final symbolic identity.*

Remark 5.4 (Gauge freedom and the choice of representative). *When the linear system for (D, K) has a nullspace, the completed block is not unique: any solution differs from another by ε -form-preserving redefinitions. All representatives are correct, but they differ wildly in the degrees of the rational functions they propagate into later rows — and degree growth compounds row by row. The production rule fixes the representative by pinning the coefficients of the highest allowed numerator monomials to zero (leaving the constant dlog residues free), which empirically selects the low-degree solution. This is a pure engineering choice invisible in the final certified ε -form, but it decides whether a family solves in minutes or in an hour.*

5.4 The final constant transformation

After the block completion, the connection is a dlog form whose residue matrices may still depend on ε (rational functions of ε times dlogs; the letters themselves are ε -free). The last step is a transformation $T(\varepsilon)$ that is *constant in the kinematics*: then $dT = 0$ and (22) is a pure conjugation $A' = T^{-1}AT$, acting letter by letter on the residues. One asks for $T(\varepsilon)$ with

$$T^{-1}(\varepsilon) \tilde{R}_a(\varepsilon) T(\varepsilon) = \varepsilon R_a \quad \text{simultaneously for all } a, \quad (31)$$

a finite-dimensional problem solved by sampling the connection at a few exact rational kinematic points and factoring the ε -dependence of the resulting constant matrices; the found $T(\varepsilon)$ is then verified by the exact symbolic identity on the full connection. Why a constant T can suffice: the offending ε -dependence originates from ε -dependent normalizations of the master basis (each sector's masters were normalized with ε -dependent prefactors), i.e. from a basis choice — and basis choices are undone by constant matrices.

6 Square roots, charts, and the Galois structure of the roots

6.1 Where the roots come from

Nothing so far required the letters to be polynomials in (v, w) — and for some blocks they are not. The origin is the geometry of the 3-particle phase space. For three massless final-state momenta with fixed invariants, the physical region is bounded by the vanishing of a Gram determinant, which in our variables is the Källén polynomial

$$\lambda_1(v, w) = (1 - v - w)^2 - 4vw. \quad (32)$$

(The name: for the triangle function $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$ one has $\lambda_1(v, w) = \lambda(1, v, w)$.) Masters whose sectors probe this boundary satisfy equations whose solutions have branch points on $\lambda_1 = 0$ — and since *square-root* branch points cannot be produced by dlogs of polynomials, their alphabets necessarily contain the algebraic letter $\sqrt{\lambda_1}$, typically through odd combinations like $d \log \frac{1-v-w-\sqrt{\lambda_1}}{1-v-w+\sqrt{\lambda_1}}$. The double-real inventory realizes five independent radicands:

$$\lambda_1 = (1-v-w)^2 - 4vw, \quad \lambda_2 = \lambda_1(-v, w), \quad \lambda_3 = \lambda_1(v, -w), \quad 4v + w^2, \quad v^2 + 4w, \quad 1 - 4vw \quad (33)$$

(six expressions; λ_2/λ_3 and the two quartic-type ones are exchanged by $v \leftrightarrow w$).

Definition 6.1 (Number of roots of a family). *The root set of a family (or block) is the set of independent square classes among the radicands appearing in its alphabet — radicands counted modulo perfect-square factors and modulo multiplicative combinations (e.g. $\sqrt{q_1}$, $\sqrt{q_2}$, $\sqrt{q_1 q_2}$ count as two roots, because the third is a product of the others). “A two-root family” means its alphabet involves two independent such roots. The inventory: 31 one-root families, 13 two-root, 3 three-root (CF259, CF300, CF303); the remaining 44 are root-free (“zero-root”).*

The root count is *measured*, not guessed: radicands are read off the derived connection and reduced to square classes exactly. It is the single number that decides which solving technology applies.

6.2 Charts: rationalizing one root

Definition 6.2 (Chart). A chart is a rational change of variables $(v, w) = (f(x, y), g(x, y))$ with non-degenerate Jacobian, such that a designated radicand becomes a perfect square: $q(f, g) = r(x, y)^2$ with r rational. In the chart, the root is the rational function r , and a block whose alphabet involves only $\sqrt{q}$ acquires a rational ε -form in (x, y) .

Example 6.3 (Deriving the Källén chart). Seek $(v, w) = (f, g)$ with $\lambda_1(f, g)$ a square. Try the product ansatz suggested by the factorized boundary:

$$v = xy, \quad w = (1 - x)(1 - y). \quad (34)$$

Then

$$1 - v - w = 1 - xy - (1 - x)(1 - y) = x + y - 2xy, \quad (35)$$

$$\lambda_1 = (x + y - 2xy)^2 - 4xy(1 - x)(1 - y) = x^2 + y^2 - 2xy = (x - y)^2. \quad (36)$$

So in this chart $\sqrt{\lambda_1} = x - y$, the Jacobian determinant is $\partial(v, w)/\partial(x, y) = x - y$ (nonzero away from the diagonal), and every odd-root letter becomes rational, e.g.

$$\frac{1 - v - w - \sqrt{\lambda_1}}{1 - v - w + \sqrt{\lambda_1}} = \frac{(x + y - 2xy) - (x - y)}{(x + y - 2xy) + (x - y)} = \frac{y(1 - x)}{x(1 - y)} \cdot \frac{2}{2} \quad \text{— a ratio of polynomial letters.} \quad (37)$$

This is the chart **Kallen1**; **Kallen2/3** are the sign variants $v = \pm xy$, $w = \mp(1 - x)(1 - y)$ rationalizing $\lambda_{2,3}$; **Q4a/b** handle $4v + w^2$ and $v^2 + 4w$ by conic parametrization; **Bilinear115** handles $1 - 4vw$.

The systematic origin of such charts: $q(v, w) = r^2$ defines a conic (in suitable projective coordinates), and a conic with one rational point is rationally parametrized by the pencil of lines through that point — the same construction that parametrizes Pythagorean triples. The catalog charts are precomputed instances of this construction, stored with their substitution, root image, Jacobian, and an exact verification that $q \circ \text{subst} = r^2$.

Working in a chart costs one *pull-back* of the whole family connection by the chain rule,

$$A_x = A_v \partial_x f + A_w \partial_x g, \quad A_y = A_v \partial_y f + A_w \partial_y g, \quad (38)$$

after which everything of Sections 4–5 applies verbatim in (x, y) , with flatness re-certified in the chart.

6.3 Two roots: joint charts

A family whose alphabet involves *two* independent roots needs a chart rationalizing both simultaneously. The construction is iterative: rationalize the first root by its catalog chart; the second radicand pulls back to a new polynomial $q_2(x, y)$, which defines a second conic *in the chart variables*; find a rational point on it and parametrize by lines again. The result is the joint charts **Kallen12**, **Kallen13**, **Kallen23** used by the thirteen two-root families. Concretely **Kallen12** keeps y and replaces x by a rational function $x(y, s)$ of a line parameter s , chosen so both λ_1 and λ_2 pull back to squares. This is why the certified two-root records are written in variables like (y, s) : they live in a joint chart.

6.4 Three roots: the Galois algebra

For the three-root families no joint chart exists — provably. The three radicands define a triple cover whose simultaneous rationalization would require the associated surface (the fibre product of the three conic double covers) to be rational; for the root triples of CF300-type families that surface is a K3 surface, which admits no rational parametrization. The obstruction is geometric, not a failure of search.

The replacement is to stop parametrizing and work in the function *field extension* directly. Adjoin the roots as symbols:

$$E = \mathbb{Q}(v, w, \varepsilon) [r_1, r_2, r_3] / (r_i^2 - q_i), \quad (39)$$

an eight-dimensional algebra with basis $\{r_S = \prod_{i \in S} r_i\}$ over the subsets $S \subseteq \{1, 2, 3\}$. Its *Galois group* is $(\mathbb{Z}/2)^3$, acting by the sign flips $r_i \rightarrow -r_i$; every function of the roots splits into eight *grades* (the characters of the group), multiplication is grade-wise ($r_S r_T = \prod_{i \in S \cap T} q_i r_{S \Delta T}$), and differentiation obeys $\partial r_S = \frac{1}{2} r_S \sum_{i \in S} \partial \log q_i$. This is “the Galois stuff”: not a technique bolted on, but the statement that the solution space organizes into representations of the root sign group, and all linear algebra (the strip equation, the modular sampling, the reconstruction) can be carried out on the eight-component coefficient vectors over the *rational* base field — with the eight sign branches recombined at the end as a consistency certificate. Modular evaluation picks primes and points where each q_i is a nonzero quadratic residue, so all eight embeddings exist over $\mathbb{F}_p$ and the grade decomposition is an honest discrete Fourier transform over $(\mathbb{Z}/2)^3$. The three-root families are the current frontier: the machinery (algebra, strip assembly, modular solving, branch certificates) is built and certified on synthetic cases; the physical blocks of CF300’s deepest sector are where it is being brought to bear.

Remark 6.4 (Roots, charts, and the involution). *The class equivalence of Definition 3.3 interacts with charts: the swap $v \leftrightarrow w$ maps $\lambda_2 \leftrightarrow \lambda_3$ and $4v+w^2 \leftrightarrow v^2+4w$, so one chart per swap-pair suffices, and inside the Källén chart the swap is the simple involution $(x, y) \rightarrow (1-x, 1-y)$.*

7 Integrating the system: transport and polylogarithms

7.1 Words along paths

With the ε -form certified, Eq. (18) reduces solving to evaluating iterated integrals of dlog forms along paths from a base point. Two standard facts organize this:

Definition 7.1 (Generalized polylogarithm). *For letters that are linear in the integration parameter t (i.e. $\varphi = t - a_i$ along the path), the iterated integrals are generalized polylogarithms (GPLs)*

$$G(a_1, \dots, a_k; t) = \int_0^t \frac{dt_1}{t_1 - a_1} G(a_2, \dots, a_k; t_1), \quad G(; t) = 1, \quad (40)$$

with weight k ; $G(\vec{0}_k; t) = \frac{1}{k!} \log^k t$, and $G(0, 1; t) = -\text{Li}_2(t)$ recovers Example 4.3.

Solving a family in practice (*transport*) means: choose a base point p_0 in the physical region (2); choose axis-aligned or otherwise convenient path segments to the point of interest; restrict the alphabet to each segment (each letter becomes a polynomial in the path parameter — segments are chosen so it is *linear*, else the segment is subdivided or re-parametrized); and accumulate the words $R_{a_k} \cdots R_{a_1} G(\dots)$ order by order to the required ε -depth. Path independence within

the region is guaranteed by flatness; crossing a letter's zero locus is forbidden (that is analytic continuation, handled deliberately, not by accident).

Two project-specific disciplines are worth knowing:

- **Depth budget.** The ε -depth to which a given master must be expanded is not uniform: a master multiplying a $1/\varepsilon^2$ coefficient in the assembled cross section needs two more orders than one multiplying a finite coefficient. The required depth per master is computed *before* transport by a reachability analysis through the assembly (of the double-real inventory, 203 masters need only the finite part, 131 one pole deeper, 9 deeper still), and transport carries exactly that depth — weight- k words are expensive, and the budget is the difference between hours and days.
- **Chart families transport in their chart.** A family whose ε -form lives in a chart is transported in the chart variables along paths chosen so all pulled-back letters stay linear in the path parameter; the result is pulled back to (v, w) only at the evaluation points.

8 Boundary conditions

8.1 The counting problem

The general solution (18) leaves one constant vector $\vec{F}(p_0) = \sum_k \varepsilon^k \vec{C}^{(k)}$ per family — naively 347 unknown constants per ε -order for the whole inventory. The first principle of the boundary program is that almost all of them are *not new numbers*: they are determined by consistency, by physics, or by constants already fixed lower in the lattice. Only what survives all constraints requires computing an integral, and each such number is a *boundary period*: the value of a cut phase-space integral in an ordered limit — a $\mathbb{Q}$ -linear combination of transcendental constants (ζ_2 , ζ_3 , $\log 2 \cdot \zeta_2$, ...) at the relevant weight.

The constraints, in the order they are applied:

1. **Normalization masters (volume rows).** The simplest master of the simplest family is the pure phase-space volume, known in closed form (next example); its value seeds the lattice, and the registry maps it into every family where that sector reappears.
2. **Inherited rows.** Families are solved in dependency order. A subsector master solved in family F_1 is (by the momentum-shift identification) a master of family F_2 's subsector; its already-known expansion becomes linear constraints on F_2 's boundary vector.
3. **Forbidden-mode (regularity) rows.** Near a singular locus the local solutions behave as powers $u^{n+a\varepsilon}$ (Section 9). The integral *representation* dictates which exponents are allowed: a cut phase-space integral has a definite power counting at $w \rightarrow 0$, at $v \rightarrow 0$, and on the soft surface $1 - v - w \rightarrow 0$, and any solution mode with an exponent outside the allowed set must have coefficient zero. Each forbidden mode is one homogeneous linear condition on the boundary vector. This is the workhorse: most constants die here.
4. **Regularity at spurious points.** Zero loci of letters that lie *inside* the physical region of a given integral (or of a subintegral) are spurious singularities for it; finiteness there kills the corresponding local modes.

Definition 8.1 (Nullity of a block). *Assemble all constraints on a block's boundary vector into a matrix C (rows: constraints; columns: local boundary modes). The number of genuinely new periods of the block is $N_{\text{new}} = \dim \ker C$, computable before evaluating anything. Blocks are processed in*

lattice order and periods deduplicated across blocks by sector identity; for the double-real inventory this leaves a period budget of a few dozen candidate periods, of which the confirmed genuinely new tier is smaller still — each then established by a dedicated exact computation (Mellin–Barnes, regions, hypergeometric expansion), with an independent numeric check that guides but never proves.

Example 8.2 (The seed: massless phase-space volume). *The n -particle massless phase-space volume in d dimensions, $\Phi_n(q^2)$, is fixed by Lorentz invariance and scaling up to the overall constant, which a short recursive computation gives as*

$$\Phi_n(q^2) = (q^2)^{(n-1)(d-2)/2 - (norm.)} \frac{(2\pi)^{\cdots} \Gamma(d/2 - 1)^{n-1} \Gamma(d-2)^{\cdots}}{\Gamma((n-1)(d-2)/2) \Gamma(n(d-2)/2)} \cdot (\text{scheme factors}), \quad (41)$$

i.e. a pure ratio of Γ -functions of ε — for the present purpose the only facts that matter are: (i) it is the $a_i=1$, numerator-free master of the lowest sector; (ii) it is computable in closed form directly from the definition, with no differential equation; (iii) expanded in ε it delivers exact ζ -values that propagate upward through the inherited rows. In the same spirit, the handful of genuinely new periods higher up are cut integrals evaluated at an ordered limit (e.g. the corner $v, w \rightarrow 0$ taken in a definite order), where they reduce to Γ -function integrals or one-fold Mellin–Barnes representations.

Example 8.3 (Boundary counting on Example 4.3). *The 2×2 system had a two-component boundary vector per ε -order. Constraints: (i) at $w \rightarrow 0$ the local exponents are $\{0, -\varepsilon\}$ (eigenvalues of εR_w with $R_w = \text{diag}(0, -1)$); if the physical integral behind F_2 is known to be finite and vanishing at $w = 0$, the $w^{-\varepsilon}$ mode is forbidden and one constant dies. (ii) F_1 is the volume-type subsector master, inherited. Nullity: 0 — no new period, and indeed the solution $\text{Li}_2(w)$ carried no free constant once $F_2(0) = 0$ was imposed. This miniature is faithfully the shape of the real bookkeeping.*

8.2 Where forbidden modes come from: a power-counting derivation

The forbidden-mode rows deserve a derivation, because they carry most of the weight. Consider a master defined as a cut phase-space integral and examine the collinear limit $w \rightarrow 0$. In the phase-space parametrization the limit corresponds to a corner where the measure and the propagators scale with definite powers of w : schematically, if the measure contributes $w^{\alpha-\beta\varepsilon} dw$ and the integrand’s propagators contribute a bounded set of extra integer powers, then the integral near the boundary is of the form

$$I(w) = \int^w dw' (w')^{\alpha-\beta\varepsilon} \rho(w') \quad \implies \quad I(w) \sim \sum_{n \geq n_0} c_n w^{n+1+\alpha-\beta\varepsilon} + (\text{analytic}), \quad (42)$$

with α, β fixed integers determined by the dimension $d = 4 - 2\varepsilon$ and the kinematics of the collinear configuration — *not* by the details of the sector. The representation therefore admits local exponents only from the arithmetic progression $\{n + a\varepsilon : a \in \mathcal{A}_{\text{phys}}\}$ with a short, computable list $\mathcal{A}_{\text{phys}}$ of allowed ε -slopes. On the other side, the differential equation offers the modes (44) with slopes from the eigenvalues of the residue matrix. Whenever the equation offers a slope the representation forbids, the coefficient of that entire mode must vanish: one exact homogeneous condition per forbidden slope per master. The same argument runs on the second collinear edge $v \rightarrow 0$ and on the soft surface $1 - v - w \rightarrow 0$ (where the scaling variable is the distance to the surface). Two practical warnings are baked into the production rules: the allowed list is computed

per chart — transplanting the (v, w) soft table into a chart frame is wrong because the chart Jacobian shifts the measure powers; and inequivalent Jordan blocks can hide a forbidden slope inside a log-degenerate mode, which is why the substrate records eigenvalues *and* Jordan data per stratum.

8.3 How many independent boundaries are there?

The question “how many independent ones?” thus has a layered answer: 347 masters $\times$ depth constants *a priori*; after volume seeding, inheritance, forbidden modes, and regularity, the kernel dimensions sum to a few dozen *candidate* periods inventory-wide; after cross-block deduplication the genuinely new tier is the short list actually computed. The discipline that keeps this honest: the nullity is computed and recorded *before* any evaluation, so “how many numbers must we produce” is a theorem about the constraint matrix, not a hope.

9 Singular structure: reading the matrix at the endpoints

9.1 Why the singularity is “in the matrix”

The connection’s poles are the letters’ zero loci. Near a locus, say $w \rightarrow 0$ with letter w , the ε -form reads

$$\partial_w \vec{F} = \frac{\varepsilon R_w}{w} \vec{F} + (\text{regular}), \quad (43)$$

whose local (Frobenius) solutions are

$$\vec{F} \sim w^{\varepsilon R_w} \vec{h}(w) = \sum_{\text{modes}} w^{n+a\varepsilon} (\log w)^j (\text{const} + O(w)), \quad (44)$$

with the exponents a the eigenvalues of R_w , integer offsets n from the expansion of the regular part, and logs from Jordan blocks. *This is the answer to “why is the singularity in the matrix”:* the residue matrix at a letter is precisely the complete catalogue of the solutions’ local behaviour there — exponents from its eigenvalues, log powers from its Jordan structure. Nothing about the endpoint behaviour needs to be discovered from the solution; it is read off the connection, exactly, before any integration.

9.2 Recovering the singular expansion, and why it must not be expanded away

Two complementary uses:

(i) Matching. The global solution (18) is a combination of GPLs; near $w = 0$ each GPL has a known expansion in $w^{a\varepsilon} (\log w)^j$ obtained by matching against (44). This matching is how boundary values “at” a singular point are defined (the constant term in the mode of exponent 0), and how the forbidden-mode rows of Section 8 are imposed.

(ii) Endpoint distributions. The cross section requires integrating the masters against smooth test functions up to the endpoint $w \rightarrow 1$ (and its analogues). A mode $(1 - w)^{-1+a\varepsilon}$ integrates to

$$(1 - w)^{-1+a\varepsilon} = \frac{1}{a\varepsilon} \delta(1 - w) + \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \left[\frac{\log^k(1 - w)}{1 - w} \right]_+, \quad (45)$$

the standard plus-distribution expansion (derive it by integrating both sides against a test function $f(w)$ and splitting off $f(1)$). The crucial observation: if one first expands the *master* in ε , the left side becomes the log tower $\frac{1}{1-w} \sum_k \frac{(a\varepsilon)^k}{k!} \log^k(1-w)$, whose term-by-term integrals are divergent — *the δ -function content is gone*. The distributional identity (45) only exists for the *unexpanded* power. Hence the pipeline’s rule: masters are delivered with their exact indicial data (which modes $n + a\varepsilon$, with which constant prefactors) alongside the expanded GPL representation; the endpoint extraction consumes the mode decomposition, and only afterwards is everything expanded in ε . “Recovering the singularity” is thus a resummation-free bookkeeping step: the matrix gave the exponents, the boundary program gave the mode coefficients, and (45) turns them into distributions.

10 Certification: the epistemology of the pipeline

Every stage above pairs a *search*, which may be heuristic, numerical, modular, or downright opportunistic, with a *certificate*, which is exact and independent of how the candidate was found:

- a derived family system is accepted with an exact flatness identity;
- a class ε -form is accepted only by exact reconstruction of the stored form from constant residues (never by the shape of some intermediate output);
- a completed family is accepted by the exact gauge identity $T^{-1}AT - T^{-1}dT = \varepsilon \sum_a R_a d \log \varphi_a$ together with flatness of the result and exact constancy of the residues — checked either symbolically or by a modular certificate with reconstruction at held-out primes;
- a boundary period enters the ledger only with an exact proof chain (its defining integral, variable maps, mode, exact value) *plus* an independent numeric check — and the numeric check guides but never proves;
- transported values are cross-checked against independent lower-depth recomputations and high-precision numerics at interior points.

The reason for the discipline is practical: with $O(10^3)$ blocks and $O(10^2)$ families, any silent wrong answer would be fatally expensive to locate later. The certificates make the final object — 347 masters as certified ε -forms plus boundary data — a self-verifying artifact.

11 The pipeline at a glance, and a glossary

Stage	Input $\rightarrow$ output	Where explained
IBP	integrals $\rightarrow$ 347 masters, 91 families	§2
DE derivation	masters $\rightarrow (A_v, A_w)$ per family, flat	§2
Blocking	family $\rightarrow$ 1119 diagonal blocks	§3
Classification	blocks $\rightarrow$ 173 classes (perm. $\circ v \leftrightarrow w$)	§3
Class ε -forms	class $\rightarrow$ canonical form, once	§5.1, §6
Completion	family $\rightarrow$ full ε -form (strips, const. $T(\varepsilon)$)	§5.2–5.4
Charts	roots $\rightarrow$ rational variables (or Galois algebra)	§6
Transport	ε -form $\rightarrow$ GPL words to budgeted depth	§7
Boundaries	constraints $\rightarrow$ nullity $\rightarrow$ few periods	§8
Endpoint	indicial data $\rightarrow \delta$ and plus distributions	§9

Glossary.**Master integral**

Basis element of the $\mathbb{Q}(v, w, \varepsilon)$ -vector space of a family's integrals modulo IBP.

Family / canonical family

Propagator set closed under scalar products; canonical = representative modulo momentum shifts ($GL(2, \mathbb{Z})$, cut-preserving), named **CFn**.

Sector

Subset of propagators actually present (positive powers); subsectors arise by deleting lines.

Block

Strongly connected group of masters in one family's DE; a diagonal block of the triangular connection.

Class

Equivalence class of blocks under basis permutation optionally composed with $v \leftrightarrow w$; 173 in this project.

Connection

The matrix one-form A in $d\vec{I} = A\vec{I}$.

Flatness

$dA = A \wedge A$; integrability of the pair (A_v, A_w) .

 ε -form / canonical form

$A = \varepsilon \sum_a R_a d \log \varphi_a$ with constant R_a .

Letter / alphabet

The functions φ_a / their set.

Residue matrix

The constant R_a ; eigenvalues give local exponents at $\varphi_a = 0$.

Gauge transformation

Basis change $\vec{I} = T\vec{F}$ acting on A by $T^{-1}AT - T^{-1}dT$.

Strip

One off-diagonal block row being completed; its unknown gauge D obeys the linear equation (28).

Root

Independent square class $\sqrt{q(v, w)}$ occurring in an alphabet.

Chart

Rational substitution making a root (or root set) rational; Kallen1/2/3, Q4a/b, Bilinear115, joint Kallen12/13/23.

Galois / multiquadratic algebra

$\mathbb{Q}(v, w, \varepsilon)[r_1..r_k]/(r_i^2 - q_i)$ with sign group $(\mathbb{Z}/2)^k$; the three-root technology.

Transport

Building the path-ordered solution as GPL words to a budgeted ε -depth.

Boundary period

Genuinely new constant surviving all boundary constraints; a cut phase-space integral at an ordered limit.

Nullity

$\dim \ker C$ of a block's boundary-constraint matrix = number of new periods, computed before evaluation.

Indicial data / modes

The exponents $n + a\varepsilon$ and log powers at a singular locus, read from the residue matrices.

11.1 One family end to end

To fix the whole chain in the mind, here is the biography of the real family CF12 (14 masters, root-free), compressed to its decision points:

1. **Reduction** named it CF12 (a canonical family after momentum-shift merging) and produced its 14 masters.
2. **DE derivation** built (A_v, A_w) by differentiating each master and IBP-reducing; the exact flatness identity certified the pair. Ordered by sectors the connection is block-triangular with 9 diagonal blocks.
3. **Classification** matched each block, by exact permutation-plus-swap identity, to one of the 173 class representatives; CF12 introduced no new class.
4. **Assembly + completion**: the stored class ε -forms were composed in (as $T_{\text{blockdiag}}$); the finite-field strip solver completed the off-diagonal blocks row by row (many with the trivial $D = 0$: their couplings were already constant-residue dlogs); a final constant $T(\varepsilon)$ concentrated the regulator; the family certificate re-derived the whole gauge identity exactly. Resulting alphabet: $\{v, w, 1-v, 1-w, v+w, 1-v-w\}$.
5. **Transport** assembled the masters as GPL words in those six letters to each master's budgeted depth.
6. **Boundary**: volume seeding through its lowest sector, inheritance from subsector families, and forbidden modes at $v \rightarrow 0$, $w \rightarrow 0$, $1-v-w \rightarrow 0$ left nullity zero — no new period; the family is fully determined.
7. **Endpoint**: its indicial data at $w \rightarrow 1$ -type loci ride along with the GPL representation for the later distribution extraction.

A two-root family (say CF231) tells the same story with two extra chapters: its connection is pulled into the joint chart Kallen23 before step 4, and its completion meets genuinely hard strips whose ansatz spans thousands of unknowns — which is exactly where the modular linear algebra earns its keep. A three-root family replaces the chart chapter with the Galois algebra of Section 6.4, and its story is still being written.

Standard references for the ideas (the notes above are self-contained, but these are the canonical entry points): Kotikov; Gehrmann–Remiddi (differential equations for masters); Henn (canonical forms); Lee (reduction algorithms; the Libra implementation); Meyer (CANONICA); Anastasiou–Melnikov (reverse unitarity); Goncharov, Duhr (polylogarithms and symbols); Besier–Wasser–Weinzierl (rationalization of square roots); Pak (canonical family mapping).